Phase diagram of networks grown with triadic closure rules

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(Dated: October 19, 2026)

We study a network that grows by three independent processes: nodes arrive at rate α and attach to a random node, open triads close at rate β_+, and closed triads open at rate β_-. Since $\alpha+\beta_++\beta_-$ only sets the clock, the model has two parameters. We derive and characterize its phase diagram. The mean degree is $\langle k\rangle = 2[1+(\beta_+-\beta_-)/\alpha]$ for $\beta_+ > \beta_-$. At $\beta_+ = \beta_-$ the triangles become a critical queue, and for $\beta_- \geq \beta_+$ the network is a tree whose degrees stay bounded, because every closure that is reversed strips a link from the centre of the triad. For $\beta_+ > \beta_-$ the outcome depends on how the open triad is found. If it is drawn uniformly from the whole network, the closure links condense onto a vanishing fraction of the nodes. Without opening, the condensate is a near-clique of order $\sqrt{n}$ nodes. It appears beyond an onset size that grows roughly as u^{-5}, $u = \beta_+/(\alpha+\beta_+)$, with no finite threshold down to $u = 0.08$. Opening thins it into a sparse club of n^a nodes, $a < 1$, which still carries every net closure link. Above $\beta_-/\beta_+ \approx 0.83$ the club dissolves, leaving a hub-free network with triangles. Local versions of the closure — a node introducing two of its friends, or befriending a friend of a friend — never condense, and the friend-of-a-friend rule reproduces the local-search model $LS(n,1)$. The two kinds of closure organize the network differently. Local closure produces assortative, hierarchical community structure, with a number of communities that grows in proportion to n. Census closure produces a strongly assortative split between a club and a periphery that barely link to each other. The tree phase is weakly disassortative and has no community structure.

I. INTRODUCTION

Triadic closure, a link forming between two nodes that share a neighbour, is the mechanism behind the high clustering of social networks [1–6]. In the local-search model $LS(n,\ell)$ [4, 7, 8] with $\ell = 1$ a newcomer picks a random node j, links to it and to a random neighbour of j. Every arrival closes one triangle, so arrival and closure are the same event, and no link is ever lost.

In a social system these are separate processes. People join at one rate, acquaintances are introduced to each other at another, and ties lapse at a third. We therefore let three processes run side by side: arrival at rate α, closure of an open triad at rate β_+, and opening of a closed triad at rate β_-. Opening is the reverse of closure, the transition from a link back to a potential link that Ref. [8], Ch. 3, sets aside. Jin, Girvan and Newman [10] combined meetings biased towards pairs with mutual friends with the decay of unrenewed ties, and found clustered networks with strong community structure. Bianconi *et al.* [11] showed that growth by triadic closure alone generates communities, and that heterogeneous node fitness replaces them with a few superhubs that attract most of the links. Here the rates are uniform, and the question is what the balance between arrival, closure and opening alone produces.

Section II defines the model and four ways of choosing the triad. Section III fixes the density and finds the first transition, at $\beta_+ = \beta_-$. Section IV gives the uncorrelated mean field. Section V lays out the phase diagram, and Secs. VI–VIII characterize its three phases. Section IX treats the local rules, Sec. X the degree correlations and communities, and Sec. XI the alternative of per-triad rates.

II. THE MODEL

The network starts from a path of three nodes. Each event is

- an *arrival*, with probability $\alpha/(\alpha+\beta_++\beta_-)$: a new node links to a uniformly random node;
- a *closure*, with probability $u_+ = \beta_+/(\alpha+\beta_++\beta_-)$: an open triad a–c–b (a, b unlinked) gets the link (a,b);
- an *opening*, with probability $u_- = \beta_-/(\alpha+\beta_++\beta_-)$: a closed triad a–c–b (a, b linked) loses the link (a,b).

How a triad is chosen defines the rule. Each closure rule has a mirror-image opening rule:

(w) *random triad*: an open (closed) triad drawn uniformly from the whole network. The centre of an open triad is then chosen with weight $\propto k_c(k_c-1)$. A link belongs to as many closed triads as it has triangles, t_e, so opening removes a link with probability $\propto t_e$;

(p) *random potential link*: closure picks a uniformly random unlinked pair with a common neighbour (the rule simulated in Ref. [8], Ch. 3); opening picks a uniformly random link with $t_e \geq 1$;

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(c) *introducer*: a random node c with $k_c \geq 2$ picks two of its neighbours and links them if they are unlinked (closure), or unlinks them if they are linked (opening);

(f) *friend of a friend*: a random node a walks $a \to j \to l$, $l \neq a$, and links to l if unlinked (closure), or unlinks from it if linked (opening).

Rules (w) and (p) need a census of the network; (c) and (f) use only a node and its neighbourhood. If the selected configuration does not qualify, the selection is repeated. A closure that finds no open triad becomes an arrival, and an opening that finds no closed triad does nothing. Opening a triad never disconnects the network, since a and b remain joined through c. Rule (w) is the main object of this paper. The reference is $LS(n,1)$, which adds one node and two links per step.

The simulation keeps adjacency lists, a hash set of links, a Fenwick tree over the triad weights $k_c(k_c-1)/2$, and, when opening is on, the triangle count t_e of every link in a second Fenwick tree. Triangle counts are updated incrementally and were checked against full recounts.

III. DENSITY AND THE FIRST TRANSITION

Write $r_\pm = \beta_\pm/\alpha$ for the number of closures and openings per arrival. Each arrival adds a node and a link, each closure adds a link, and each successful opening removes one. If every opening finds a closed triad, then after n arrivals $E = n(1 + r_+ - r_-)$ and

$$\langle k\rangle = 2(1+\Delta), \qquad \Delta = \frac{\beta_+ - \beta_-}{\alpha}. \tag{1}$$

To keep the density constant the rates must be held in fixed ratio, with $\beta_+ - \beta_- = \alpha(\langle k\rangle/2 - 1)$. Without opening, $\beta_+ = \alpha$ reproduces the density of $LS(n,1)$, $\langle k\rangle = 4$.

Equation (1) cannot hold for $\beta_- \geq \beta_+$, since it would give $\langle k\rangle \leq 2$ while the network stays connected, $E \geq n-1$. The openings must then fail often enough to keep $E \geq n-1$. Close to a tree each closure creates one triangle and each opening destroys one. The triangle count is therefore a queue with arrival rate β_+ and service rate β_-. It grows linearly in n for $\beta_+ > \beta_-$, is critical at $\beta_+ = \beta_-$, and stays $O(1)$ for $\beta_- > \beta_+$. In the last case a fraction $(\beta_- - \beta_+)/\beta_-$ of the openings find nothing to remove.

The simulations follow this exactly. At $\beta_+ = \alpha$ and $n = 10^6$ the fraction of failed openings is 0.093, 0.202, 0.501, 0.751 and 0.900 for $\beta_-/\beta_+ = 1.1$, 1.25, 2, 4, 10, against the predicted 0.091, 0.200, 0.500, 0.750, 0.900. At $\beta_+ = \beta_-$ the network holds $T = 317$ triangles at $n = 10^6$, of order $\sqrt{n}$ as for a critical queue. The excess degree $\langle k\rangle - 2 = 2\Delta$ vanishes linearly at $\beta_- = \beta_+$, and the triangles per node fall from 0.75 at $\beta_-/\beta_+ = 0.5$ to 3×10^{-4} at 1 and 10^{-5} at 1.1.

The parameter plane is therefore cut by the diagonal $\beta_- = \beta_+$. Below it the network is sparse with extensive triangles and density given by Eq. (1); above it the network is a tree. Since only ratios of rates matter, we use two coordinates: the net growth Δ, which fixes the density, and the turnover β_-/β_+.

IV. UNCORRELATED MEAN FIELD

Let a closure pick e of its two endpoints with probability proportional to degree and the other $2-e$ uniformly. In the absence of correlations, the degree of a node grows with the number of arrivals as

$$n\frac{dk_i}{dn} = 1 + r_+(2-e) + \frac{r_+ e}{\langle k\rangle}k_i - r_-\frac{2t_i}{3T/n}, \tag{2}$$

where the last term is opening under rule (w): node i is an endpoint of the removed link with probability $2t_i/3T$, with t_i its number of triangles. Rules (w), (c) and (p) reach both endpoints through a link, so $e = 2$. Rule (f) starts from a uniformly random node, so $e = 1$.

Without opening, $k_i \sim (n/n_i)^b$ with $b = e\,u/2$, $u = \beta_+/(\alpha+\beta_+)$, and

$$\gamma = 1 + \frac{1}{b} = 1 + \frac{2}{e\,u}. \tag{3}$$

The density cancels from the preference because of Eq. (1). At $u = 1/2$, rules (w), (c) and (p) give $\gamma = 3$, and rule (f) gives $\gamma = 5$, which is the exponent of $LS(n,1)$ itself [8]. In $LS(n,1)$ the newcomer is the uniform endpoint and the neighbour of j the preferential one. Separating arrival from closure leaves the exponent unchanged as long as one endpoint of the closure stays uniform.

With opening under rule (w), and clustering falling as $1/k$, we have $t_i \propto k_i$ and $\sum_i t_i = 3T$, so the opening term is $-2r_-k_i/\langle k\rangle$. Then

$$n\frac{dk_i}{dn} = 1 + \Big(1 - \frac{2}{\langle k\rangle}\Big)k_i, \tag{4}$$

and $\gamma = 1 + \langle k\rangle/(\langle k\rangle - 2)$, independent of β_- at fixed density. In this approximation turnover does nothing. All the structure below therefore comes from correlations.

V. THE PHASE DIAGRAM

To tell condensation from an ordinary heavy tail we use the *club fraction* $\phi(m)$: the fraction of all links that join two of the m highest-degree nodes. In a growing network without condensation the richest n^a nodes are essentially the oldest, and $\phi(n^a)$ decays as n^{a-1}. In a condensate, a fraction of the links sits on a vanishing fraction of the nodes, so $\phi(n^a)$ stays finite for some $a < 1$. A fraction $\Delta/(1+\Delta)$ of all links are net closures, and we normalize by it,

$$\psi = \frac{1+\Delta}{\Delta}\,\phi(n^{0.8}). \tag{5}$$

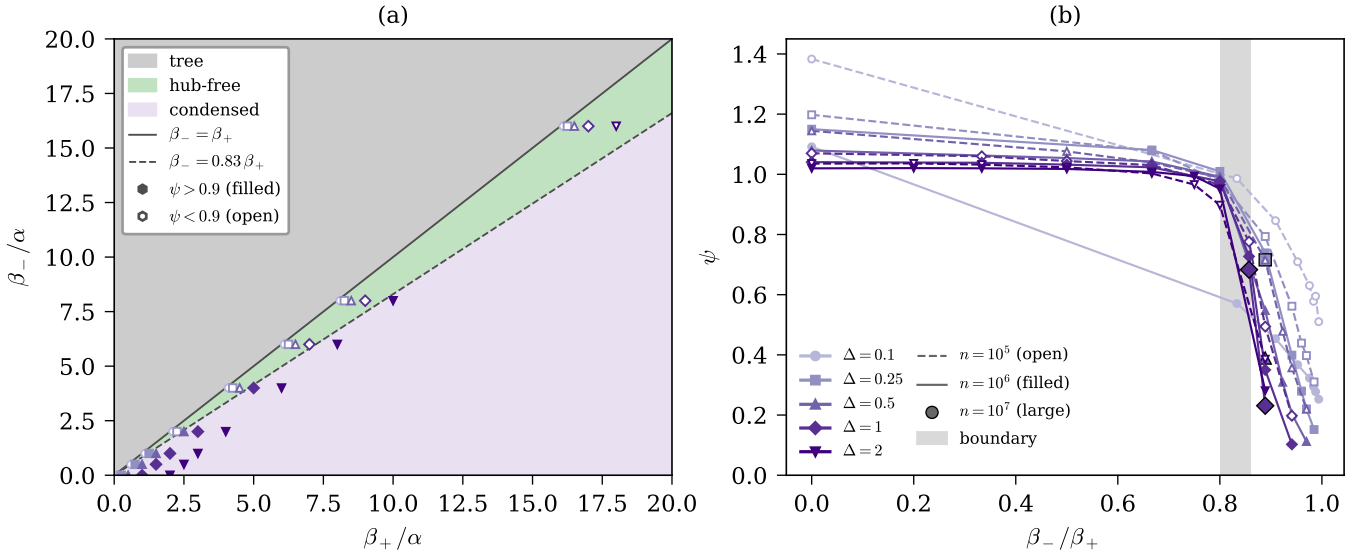


FIG. 1. Phase diagram of rule (w). (a) The plane of closure and opening rates. Shades: tree phase (grey, $\beta_- \geq \beta_+$), hub-free phase (green), condensed phase (purple). Solid line, $\beta_- = \beta_+$; dashed line, $\beta_- = 0.83\beta_+$. Symbols are simulations at $n = 10^6$ for net growth $\Delta = 0.1, 0.25, 0.5, 1, 2$ (light to dark; symbols as in (b)), filled when the club share $\psi > 0.9$. (b) Club share ψ (fraction of the net closure links held by the $n^{0.8}$ highest-degree nodes) against β_-/β_+. Symbol and colour give Δ; open symbols and dashed lines, $n = 10^5$; filled symbols and solid lines, $n = 10^6$; large symbols, $n = 10^7$. The grey band marks the boundary of the condensed phase.

$\psi \approx 1$ means that every net closure link lies inside the club of $n^{0.8}$ nodes.

We ran rule (w) at $\Delta = 0.1$, 0.25, 0.5, 1, 2 and $\beta_-/\alpha =$ 0, 0.5, 1, 2, 4, 6, 8, 16, at $n = 10^5$ and 10^6, plus four points at $n = 10^7$ (data/phase.csv). The result is Fig. 1. There are three phases:

- *condensed*, $\beta_- \lesssim 0.83\beta_+$: $\psi \approx 1$, not decreasing with n;
- *hub-free*, $0.83\beta_+ \lesssim \beta_- < \beta_+$: extensive triangles, $\langle k \rangle$ from Eq. (1), and ψ decaying with n;
- *tree*, $\beta_- \geq \beta_+$: $\langle k \rangle = 2$, $O(1)$ triangles, and bounded degrees.

Plotted against β_-/β_+, the club share collapses for all five densities (Fig. 1b). The two boundaries are therefore, to the resolution of the data, lines of constant ratio β_-/β_+. The degree distributions of the three phases are in Fig. 2b.

VI. THE CONDENSED PHASE

A. Without opening: a dense core

With $\beta_- = 0$ the only parameter is $u = \beta_+/(\alpha + \beta_+)$, and $\langle k \rangle = 2/(1-u)$. Under rule (w) at $u = 1/2$, $n = 10^6$, the $10^3 = \sqrt{n}$ highest-degree nodes are linked to one another with density 0.81 and hold 20% of all links. The 100 richest form a complete graph, and all of them were born among the first few thousand arrivals. This share does not shrink with size: $\phi(\sqrt{n}) = 0.176$, 0.192, 0.202 at $n = 10^4$, 10^5, 10^6, while the density climbs to 0.71, 0.77, 0.81. The same holds for every $u \geq 0.3$. The core is a dense block of $\Theta(\sqrt{n})$ nodes inside a sparse network. The number of triangles therefore grows as $T \sim n^{3/2}$, and the transitivity stays near 0.5 while the mean local clustering vanishes (Table I). The degree distribution has a second maximum at $k \sim \sqrt{n}$ (Fig. 2a), and the degree of the oldest nodes saturates with age.

The hub degree follows from the core's own triads. Once triads centred in the core dominate the census, a core node a is an endpoint of a closure with probability w_a/W. Here $w_a \simeq \rho\sqrt{n}\,K$ counts the open triads through its $\rho\sqrt{n}$ core neighbours, and $W \simeq \sqrt{n}\,K^2/2$. Hence $n\,dK/dn \simeq 2\rho r_+ n/K$ and

$$K^2 \simeq c\, r_+\, n, \tag{6}$$

TABLE I. Rules without opening at $u = 1/2$ ($\langle k \rangle = 4$), $n = 10^6$. b: age exponent, fitted over $10^2 < n_i < 10^5$; $b_{\rm MF} = eu/2$; $\mathcal{T}$: transitivity; $\bar{C}$: mean local clustering; r_k: degree assortativity; $\phi(\sqrt{n})$: club fraction of the $\sqrt{n}$ highest-degree nodes; ρ: link density among them.

rule	b	$b_{\rm MF}$	$k_{\max}$	$\mathcal{T}$	$\bar{C}$	r_k	$\phi(\sqrt{n})$ (ρ)
$LS(n,1)$	0.27	1/4	169	0.25	0.69	0.05	0.001 (0.002)
(f)	0.31	1/4	203	0.24	0.37	0.28	0.002 (0.008)
(c)	0.37	1/2	391	0.22	0.42	0.16	0.002 (0.007)
(p)	0.66	1/2	684	0.03	0.02	0.52	0.026 (0.11)
(w)	0.89	1/2	1619	0.48	0.008	0.65	0.20 (0.81)

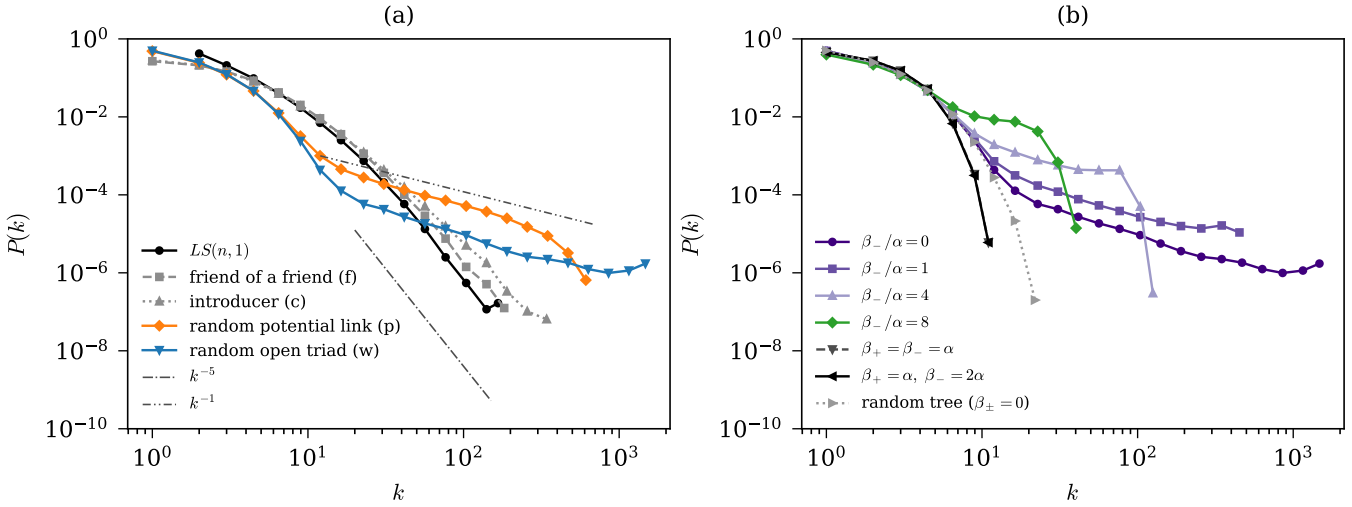


FIG. 2. Degree distributions in logarithmic bins, $n = 10^6$. (a) The closure rules without opening at $\langle k\rangle = 4$ ($u = 1/2$). Reference lines, k^{-5} and k^{-1}. (b) Rule (w) across the phase diagram: at $\langle k\rangle = 4$ with increasing turnover, $\beta_-/\alpha = 0, 1, 4$ (condensed, purple) and 8 (hub-free, green); at the tree transition $\beta_+ = \beta_- = \alpha$ (dashed); in the tree phase $\beta_+ = \alpha$, $\beta_- = 2\alpha$ (black); and the uniform random tree, $\beta_\pm = 0$ (dotted).

with c of order 4ρ. The data give $k_{\max}^2/(r_+ n) = 2.8$, 2.7, 2.6, 2.5 at $u = 0.3$–0.6, $n = 10^6$, drifting down by about 10% per decade.

Onset. At small u a long normal plateau precedes the condensate. Along it T/n is constant and $k_{\max}/\sqrt{n}$ falls. The plateau ends abruptly: T/n turns up, and $k_{\max}/\sqrt{n}$ rises back to its condensed value (Fig. 3a). Taking the onset size n^* as the first n with $T/n \geq 1$, we find

u	0.08	0.10	0.12	0.15	0.20	0.25
$\ln n^*$	15.4	14.5	13.3	12.4	11.1	9.4

with seed-to-seed spread 0.1–0.5. Over $0.1 \leq u \leq 0.3$ a power law, $n^* \propto u^{-5.4}$, fits with rms 0.17 in $\ln n^*$. An essential singularity, $\ln n^* = 6.1 + 0.87/u$, fits with rms 0.41. Fitted before the $u = 0.08$ runs existed, these two forms predicted $\ln n^* = 15.7$ and 17.0 for $u = 0.08$. Two runs to 3×10^7 both condensed at 15.4, and five runs stopped at 3×10^6 had not condensed yet, as both forms require. At $u = 0.05$ nothing happens up to 10^7; the power law puts the onset at 8×10^7. Plotted against $1/u$, the onset curve bends down (Fig. 3b). A finite threshold $u_c > 0$, where n^* would diverge, would bend it up. The data thus favour condensation at every $u > 0$, with an onset size that becomes astronomical as $u \to 0$. A threshold below $u = 0.08$ cannot be excluded.

Mechanism. Under rule (w) a node is an endpoint of a closure with probability w_a/W, where $w_a = \sum_{c\in\partial a}(k_c - 1) - 2t_a$ counts the open triads ending at a. This rate is set by the degrees of a's neighbours, not by its own. In the uncorrelated limit $w_a \propto k_a$, which gives Eq. (2). But a hub of degree K is the centre of $\sim K^2/2$ triads, and each closure centred on it raises the degree of two of its neighbours. A two-variable caricature captures the feedback. Let s be the summed degree of the hub's neighbours, $x = \ln n$, and $\epsilon = r_+/\bar{w}$, with $\bar{w}$ the mean number of open triads per node. Then $K' \simeq \epsilon s$ and $s' \gtrsim \epsilon K^2$, so $K'' \simeq \epsilon^2 K^2$. This blows up at a finite $x^* \sim \epsilon^{-1} K_0^{-1/2}$ for any $\epsilon > 0$, so the caricature has no threshold either. The blow-up stops only when the core dominates W, which gives Eq. (6). A symptom of the feedback is visible before the onset: in the normal plateau the five largest hubs keep a constant local clustering of ≈ 0.1 for every u between 0.02 and 0.1, instead of the $C \sim 1/k$ hierarchy of the local rules.

Rule (p). The potential-link rule also concentrates links on the oldest nodes (Fig. 2a). The 1% richest nodes hold 6.5%, 18% and 37% of all links at $n = 10^4$, 10^5, 10^6 ($u = 1/2$). Its core is sparse, with density 0.1 among the $\sqrt{n}$ richest nodes and transitivity 0.03–0.017. Its largest degree grows as about $n^{0.35}$. Without the m-fold weight of pairs sharing m neighbours there is no drive towards a clique, but the links still concentrate. The mean-field exponent of Ref. [8] for this rule ($\gamma = 2.62$ at $u = 1/2$) assumes a closure rate per potential link. That makes the total closure rate proportional to the number of potential links, which is not the rule simulated, and which does not conserve the density (Sec. XI).

B. With opening: a sparse club

Opening hits a link in proportion to its triangles, so it falls mainly on the core. The core thins out (Fig. 4). At $\langle k\rangle = 4$ and $n = 10^6$, the share of links among the $\sqrt{n}$ richest nodes falls from 0.20 to 0.09, 0.02, 0.005 and 0.001 at $\beta_-/\alpha = 0$, 0.5, 1, 2, 4. The links have not left the rich nodes, however. The $n^{0.8}$ richest nodes hold 0.52, 0.52, 0.52, 0.51, 0.49 of all links: essentially every

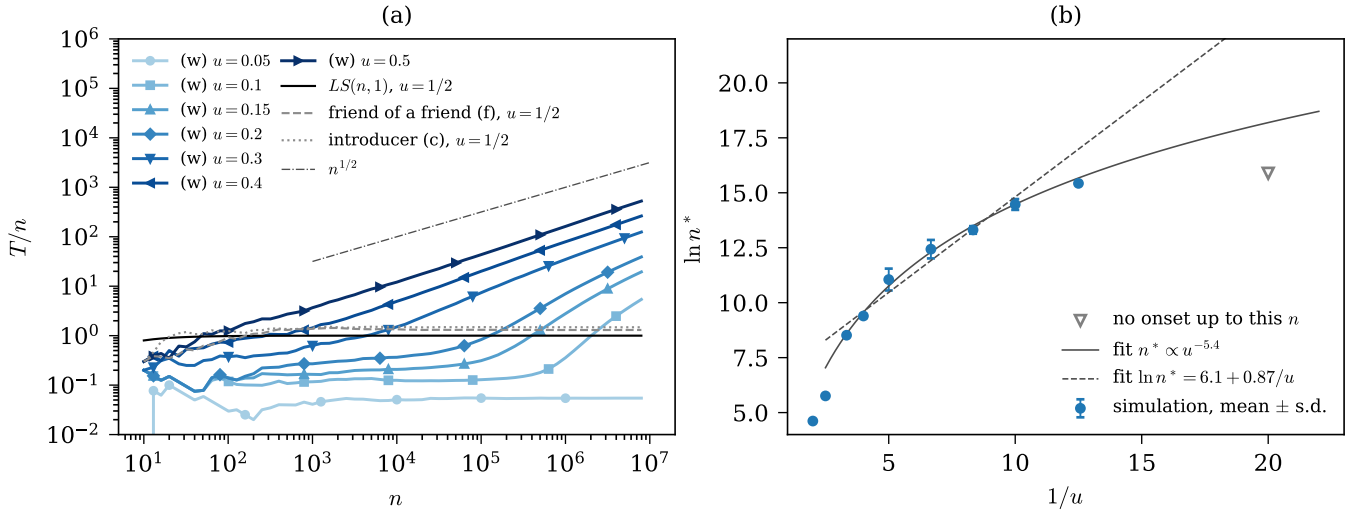


FIG. 3. Condensation without opening. (a) Triangles per node T/n along single trajectories to $n = 10^7$: rule (w) at $u = 0.05$–0.5 (blue lines with symbols, light to dark), and the local rules and $LS(n, 1)$ at $u = 1/2$ (grey and black). Dash-dotted line, $n^{1/2}$. (b) Onset size n^*, the first n with $T/n \geq 1$, mean and standard deviation over 5–6 seeds for $0.1 \leq u \leq 0.25$ and one or two trajectories otherwise. Lines are fits over $0.1 \leq u \leq 0.3$: a power law (solid) and an essential singularity (dashed). The point at $1/u = 12.5$ ($u = 0.08$, two runs to 3×10^7) was not used in the fits. Triangle: $u = 0.05$, no onset up to $n = 10^7$.

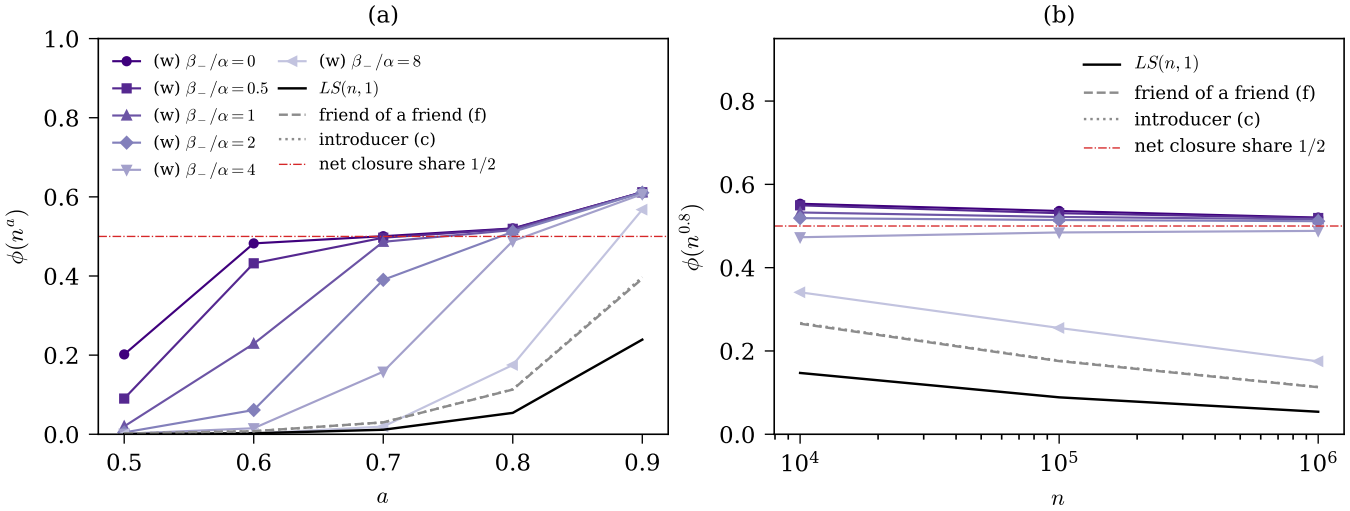


FIG. 4. The condensed phase under turnover, rule (w) at $\langle k \rangle = 4$ ($\beta_+ = \alpha + \beta_-$). (a) Club fraction $\phi(n^a)$ against the club size exponent a, $n = 10^6$, for rule (w) at $\beta_-/\alpha = 0$–8 and for $LS(n, 1)$ and the local rules without opening. Dash-dotted line: the net closure share, 1/2. (b) $\phi(n^{0.8})$ against n (mean over 4, 2, 1 seeds at $n = 10^4$, 10^5, 10^6). Purple symbols: rule (w), as in (a); grey and black lines: local rules and $LS(n, 1)$ (the dotted introducer line lies under the dashed friend-of-a-friend line).

net closure link, whose share is $\Delta/(1 + \Delta) = 1/2$. These fractions change by less than 0.04 between $n = 10^4$ and 10^6 (Fig. 4b). Under the local rules the same fraction falls as in any non-condensed network: 0.27, 0.18, 0.11 for (c) and (f), and 0.15, 0.09, 0.05 for $LS(n, 1)$.

Turnover therefore spreads the condensate into a sparse club. Define its size n^{a^*} by $\phi(n^{a^*}) = \Delta/[2(1+\Delta)]$, half the net closure links. Then a^* grows from 0.52 at $\beta_- = 0$ to 0.73 at $\beta_-/\alpha = 4$, and at fixed β_- it does not move between $n = 10^4$ and 10^6 (Fig. 4a). The largest degree follows. Between $n = 10^6$ and 8×10^6, $k_{\max} \sim n^\theta$ with $\theta = 0.52$, 0.44, 0.33, 0.23 at $\beta_-/\alpha = 0.5$, 1, 2, 4, against $\theta = 1/2$ at $\beta_- = 0$. The degree distribution loses its second maximum, and its tail ends at progressively smaller k (Fig. 2b, purple).

VII. THE HUB-FREE PHASE

At larger turnover the club dissolves. In Fig. 1b, $\psi \approx 1$ up to $\beta_-/\beta_+ = 0.8$ for every density and then drops. Every point at $\beta_-/\beta_+ = 0.8$ has ψ flat or rising between $n = 10^5$ and 10^6 (four densities; $\Delta = 0.1$ has no such point). Every point at 0.83 and above falls. Four points just above the boundary, followed to $n = 10^7$, keep falling and none turns back up. For $n = 10^5$, 10^6, 10^7: $\psi = 0.78$, 0.73, 0.68 at $\Delta = 1$, $\beta_-/\beta_+ = 0.857$; and, at $\beta_-/\beta_+ =$ 0.889, 0.79, 0.74, 0.72 at $\Delta = 0.25$; 0.72, 0.55, 0.39 at $\Delta =$ 0.5; 0.49, 0.35, 0.23 at $\Delta = 1$. We place the boundary at $\beta_-/\beta_+ = 0.80$–0.86, the same for all Δ within that resolution. Near it the decay is slow, and without opening the onset of condensation moved to very large n at small u. Sizes beyond 10^7 could still shift the boundary slightly.

In the hub-free phase the network keeps extensive triangles and the density of Eq. (1), but it has no condensate. The largest degree grows slowly (for $\Delta = 1$: $k_{\max} = 37$, 45 at $\beta_-/\alpha = 8$ and 23, 26 at 16, for $n = 10^5$, 10^6). The degree distribution falls off steeply beyond $k \approx 40$ (Fig. 2b, green), and the assortativity decreases with turnover, from 0.51 to 0.30 between $\beta_-/\alpha = 8$ and 16 at $\Delta = 1$.

VIII. THE TREE PHASE

For $\beta_- \geq \beta_+$ nearly every triangle is opened soon after it forms. One time in three the opening removes the link just added, which undoes the closure. The other two times it removes one of the two links at the centre c, whose link passes to one of c's neighbours, so c loses one link. Under rule (w) the centre is drawn with weight $k_c(k_c-1)$. A node therefore loses links at a rate quadratic in its degree and gains them, as a neighbour of a centre, at a rate linear in it. The degrees are confined.

The data confirm this (Fig. 2b). At $\beta_+ = \alpha$, the largest degree is 11–13 for every $\beta_- \geq \beta_+$, at both $n = 10^5$ and 10^6. The uniform random tree of the same size reaches 24. The distribution bends down on the log-log plot. The ratio $P(k)/P(k-1)$ falls from 0.6 at $k = 2$ to 0.2 at $k = 10$, where the uniform random tree has a constant 1/2. The assortativity is -0.14. The critical network at $\beta_+ = \beta_-$ has the same degree distribution as the tree phase. The two differ only in their $O(\sqrt{n})$ versus $O(1)$ triangles.

IX. LOCAL RULES

The local rules never condense. Without opening (Fig. 2a and Table I), they follow Eq. (3) up to corrections from correlations and a five-decade window. The age exponent b is 0.31 for (f) (mean field 1/4) and 0.37 for (c) (mean field 1/2), against 0.27 for $LS(n, 1)$ (exact 1/4). Their mean local clustering stays near 0.4, and the club fraction decays with n (Fig. 4b).

With opening at $\langle k \rangle = 4$, rule (f) keeps its age exponent at 0.31–0.34 for $\beta_-/\alpha = 0$ to 4, so it stays in the $LS(n, 1)$ class. Its mean clustering falls from 0.37 to 0.12. Rule (c) develops a broader tail ($k_{\max} = 391$ to 1287, age exponent 0.37 to 0.46), but the 1% richest nodes still hold only 2% of the links. On approaching $\beta_- = \beta_+$ at $\beta_+ = \alpha$ the hubs of rule (f) fade ($k_{\max} = 95$, 37, 17 at $\beta_-/\beta_+ = 0.5$, 0.8, 0.9), and at $\beta_+ = \beta_-$ it reaches the same critical tree ($k_{\max} = 12$, $T = 147$). For $\beta_- > \beta_+$ we did not run the local rules. Their opening starts from a random node and must find one of $O(1)$ triangles, which takes too long to simulate.

X. DEGREE CORRELATIONS AND COMMUNITIES

Assortativity. Census closure joins two nodes that share neighbours, which under rule (w) are mostly old and rich, so it makes the network assortative [13]. Over the whole condensed phase the degree assortativity is $r = 0.55$–0.68 at $n = 10^6$ for $\Delta \geq 0.25$ (Fig. 5a). It does not mark the condensation boundary. Just inside the hub-free phase r is still about 0.5, and it grows with n: 0.46, 0.51, 0.55 at $n = 10^5$, 10^6, 10^7 for $\Delta = 1$, $\beta_-/\beta_+ = 0.89$. It falls to zero only as $\beta_- \to \beta_+$, where the closure links that carry it vanish. At $\beta_+ = \alpha$,

TABLE II. Communities at $n = 10^5$ (in parentheses, $n = 10^4$). $B_{\rm pp}$: groups of the planted-partition model [16]; Q: modularity of that partition; S_1: fraction of nodes in its largest group; $\bar{k}_s$: mean degree of the nodes outside the largest group; $B_{\rm sbm}$ and L: groups at the bottom level and number of nontrivial levels of the nested degree-corrected SBM [18]. Rule (w) at $\langle k \rangle = 4$; $\beta_-/\beta_+ = 0.86$ lies on the condensation boundary, and 0.89 and 0.94 in the hub-free phase. Every configuration-model rewiring of these networks gives $B_{\rm pp} = B_{\rm sbm} = 1$.

network	$B_{\rm pp}$	Q	S_1	$\bar{k}_s$	$B_{\rm sbm}$	L
$LS(n, 1)$	1870 (216)	0.95	0.002	4.0	2435	9
(f), $u = 1/2$	1265 (154)	0.90	0.005	4.0	1863	7
(c), $u = 1/2$	1049 (148)	0.92	0.005	4.0	2128	9
(p), $u = 1/2$	3 (8)	0.43	0.94	36	3	1
(w), $\beta_-/\beta_+ = 0$	25 (2)	0.27	0.95	39	4	2
(w), $\beta_-/\beta_+ = 0.33$	26 (3)	0.28	0.94	32	4	2
(w), $\beta_-/\beta_+ = 0.5$	35 (10)	0.31	0.87	18	3	1
(w), $\beta_-/\beta_+ = 0.67$	2 (7)	0.43	0.92	29	6	3
(w), $\beta_-/\beta_+ = 0.8$	2 (2)	0.42	0.81	13	2	1
(w), $\beta_-/\beta_+ = 0.86$	3 (2)	0.41	0.73	9.4	2	1
(w), $\beta_-/\beta_+ = 0.89$	2 (2)	0.37	0.60	7.0	2	1
(w), $\beta_-/\beta_+ = 0.94$	1 (1)	0	1	—	1	0
(w), $\beta_+ = \beta_-$	1 (1)	0	1	—	1	0
(w), $\beta_- = 2\beta_+$	1 (1)	0	1	—	1	0
random tree	1 (1)	0	1	—	1	0

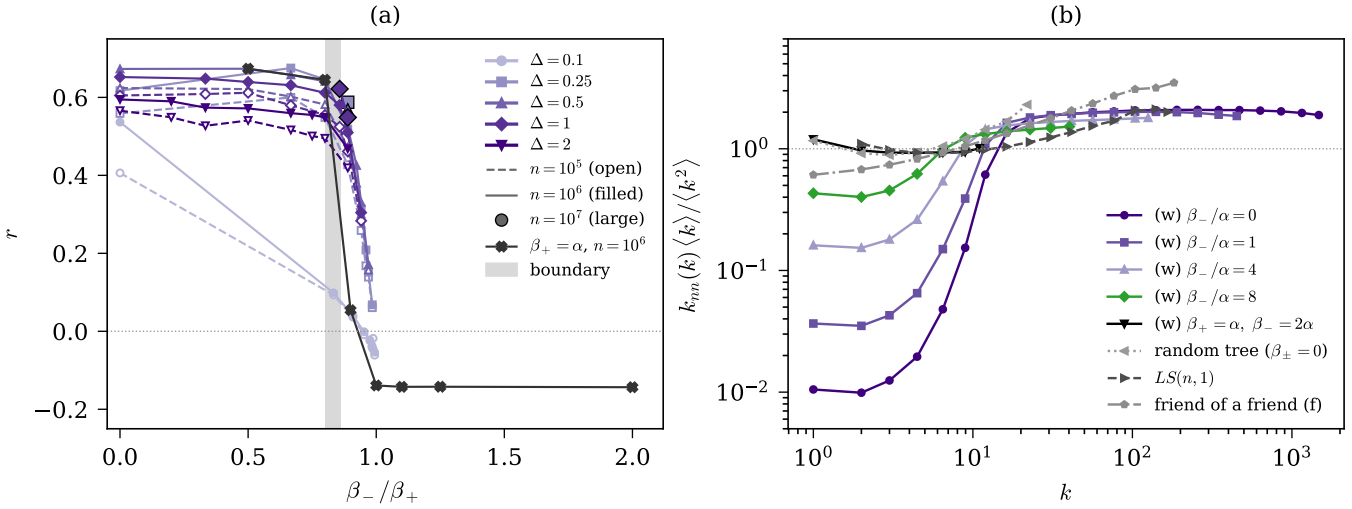


FIG. 5. Degree correlations. (a) Assortativity r of rule (w) against β_-/β_+ on the grid of Fig. 1 (symbols as in Fig. 1b), and across $\beta_- = \beta_+$ at $\beta_+ = \alpha$, $n = 10^6$ (crosses). The grey band marks the boundary of the condensed phase; dotted line, $r = 0$. (b) Mean degree of the neighbours of degree-k nodes, divided by its uncorrelated value $\langle k^2\rangle/\langle k\rangle$, in logarithmic bins, $n = 10^6$: rule (w) at $\langle k\rangle = 4$ with $\beta_-/\alpha = 0$, 1, 4 (condensed, purple) and 8 (hub-free, green), in the tree phase (black), the uniform random tree (dotted), and $LS(n, 1)$ and rule (f) at $u = 1/2$ (grey). Dotted line: no correlations.

$r = 0.67$, 0.65, 0.05 at $\beta_-/\beta_+ = 0.5$, 0.8, 0.9. In the tree phase it is $r = -0.14$ for every $\beta_- \geq \beta_+$ at both $n = 10^5$ and 10^6. The uniform random tree of the same size has $r = -0.001$. The tree phase is therefore mildly disassortative: leaves attach preferentially to the few nodes of larger degree that the pruning of triad centres leaves behind. The local rules sit in between, with $r = 0.05$–0.28 (Table I).

Neighbour degree. The mean degree $k_{nn}(k)$ of the neighbours of degree-k nodes [14] shows how the assortativity is built (Fig. 5b). Under rule (w) without opening it is a step. Nodes with $k \leq 4$ have neighbours of degree $0.01\,\langle k^2\rangle/\langle k\rangle$, about 5, and nodes with $k \gtrsim 30$ have neighbours of twice the uncorrelated degree. Between $k \approx 5$ and 30 the curve rises by two decades. The periphery is linked to the periphery and the club to the club. Turnover lowers the step: the low-k plateau rises to 0.04, 0.16 and 0.43 at $\beta_-/\alpha = 1$, 4, 8, and the high-k plateau falls from 2 to 1.4. In the tree phase $k_{nn}(k)$ falls from 1.2 at $k = 1$ to 0.93 at $k \geq 4$, close to the uniform random tree. The local rules instead give a smooth rise, from 0.6–1.1 at the smallest degrees to 2–3 at $k \approx 100$.

Communities. Triadic closure is known to generate communities [10, 11]. We look for them by statistical inference with graph-tool [15]. We fit two models: the planted-partition model [16], which describes assortative communities only and returns one group when there are none, and the nested degree-corrected stochastic block model (SBM) [17, 18], which describes any mixing pattern. Both select the number of groups by minimum description length, so, unlike modularity maximization [19], they do not find groups in random graphs. As a check, each network is rewired by the configuration model and refitted. Every rewired network gives a single group, so the structure below does not come from the degree sequence. Table II gives the results at $n = 10^5$ and 10^4.

The two kinds of closure give opposite pictures. The local rules and $LS(n, 1)$ have strong community structure. They split into about 10^3 assortative groups at $n = 10^5$ with modularity $Q = 0.90$–0.95, and the number of groups grows nearly in proportion to n, by factors of 7 to 9 between 10^4 and 10^5. The SBM nests them in 7 to 9 levels, a hierarchy of communities, as in the triadic-closure models of Refs. [10, 11].

Rule (w) has no such hierarchy. The SBM finds two to six groups at every turnover of the condensed phase, and at most three levels. One group holds 73–95% of the nodes. The others contain the rich nodes: their mean degree is 9 to 39, against $\langle k\rangle = 4$. With two groups, 93–95% of the links lie inside a group, so the club and the periphery barely link to each other. This is the k_{nn} step seen in block form. The modularity of this split, 0.4, measures a club against a periphery, not a set of communities. At $\beta_- \leq \beta_+/2$ the planted-partition fit also cuts the periphery into 25–35 groups, but only half the links fall inside them ($Q \approx 0.3$). Rule (p), whose core is sparse, gives the same club-and-periphery picture. Turnover shrinks the periphery's share, from 95% of the nodes at $\beta_- = 0$ to 60% at $\beta_-/\beta_+ = 0.89$. The split survives just above the condensation boundary at these sizes, where the club fraction is still decaying slowly (Sec. VII). Further in, at $\beta_-/\beta_+ = 0.94$, there is no structure. The critical network, the tree phase and the uniform random tree also have none: a tree with bounded degrees offers no evidence for groups.

Bianconi *et al.* [11] found that triadic closure makes

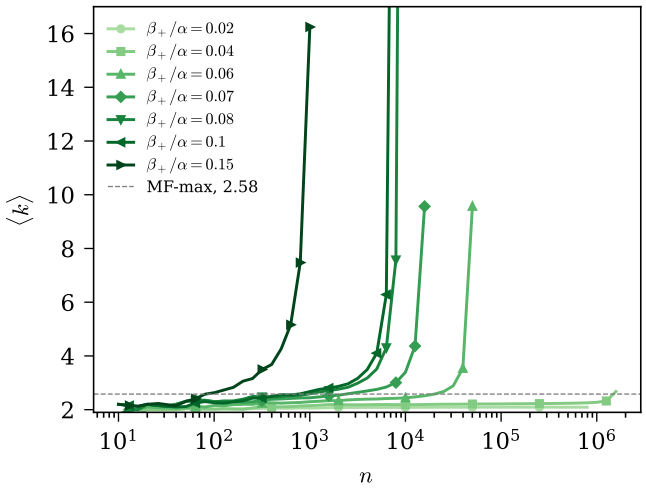


FIG. 6. Mean degree under per-triad rates, rule (w), $\beta_- = 0$ (closure probability per event $\beta_+ W/(\alpha n + \beta_+ W)$). Green lines with symbols, light to dark: $\beta_+/\alpha = 0.02$–0.15. Runs to 10^6 stop at $E/n = 50$; the run at 0.04 was continued to 1.6×10^6 and stopped inside the runaway. Dashed line (MF-max): the largest stationary mean-field density, 2.58.

communities, and that heterogeneous fitness destroys them in favour of superhubs that share the rest of the nodes. Here the same replacement happens with uniform rates, when closure draws its triads from the whole network instead of from the neighbourhood of a node.

XI. PER-TRIAD RATES

An alternative reading makes the rates per capita: every node spawns a newcomer at rate α and every open triad closes at rate β_+ (here $\beta_- = 0$). The closure rate is then $\beta_+ W$, and W/n is not a constant of the dynamics. In the uncorrelated limit $n \propto e^{\alpha t}$ and $dk_i/dt = \alpha + \beta_+ m k_i$, with $m = 2W/\sum_j k_j$. With $x = \beta_+ m/\alpha$ and age in units of $1/\alpha$,

$$k(\tau) = \left(1 + \tfrac{1}{x}\right) e^{x\tau} - \tfrac{1}{x}, \qquad \langle k \rangle = \frac{2}{1-x}, \qquad \gamma = 1 + \frac{1}{x}. \tag{7}$$

Self-consistency requires $m = \langle k(k-1) \rangle / \langle k \rangle$, which is finite only for $x < 1/2$, and fixes $\beta_+/\alpha = x/m(x)$. This function has a maximum, $\beta_+/\alpha = 0.067$ at $x = 0.22$, where $\langle k \rangle = 2.58$. Above it no stationary state exists.

Simulations (Fig. 6) show the runaway for $\beta_+/\alpha \geq 0.06$ within 10^3–5×10^4 nodes. At 0.02 and 0.04, $\langle k \rangle$ looks stationary for a while, at 2.09 and 2.23. The plateau at 0.04 corresponds to an effective $u = 1 - 2/\langle k \rangle \approx 0.10$, for which Sec. VI predicts condensation near 2×10^6. The run condenses near 10^6 and then runs away ($\langle k \rangle = 2.67$ at 1.6×10^6 and rising): once a core forms, $W \sim n^{3/2}$, and closures outpace arrivals. Since condensation appears to happen at every $u > 0$, per-triad rates should have no stationary density at any $\beta_+ > 0$. Total rates in fixed ratio are the consistent choice.

XII. DISCUSSION

The model has two parameters once the clock is fixed: the net growth $\Delta = (\beta_+ - \beta_-)/\alpha$, which sets the density, and the turnover β_-/β_+. Its phase diagram has two boundaries, both at fixed turnover.

- At $\beta_- = \beta_+$ the triangles become a critical queue. Below it they are extensive and $\langle k \rangle = 2(1 + \Delta)$. Above it the network is a tree whose degrees stay bounded, because a reversed closure strips a link from the centre of the triad, and the centre is chosen with weight k^2.

- At $\beta_-/\beta_+ \approx 0.83$ the condensate dissolves. Below it the net closure links concentrate on a club of n^a nodes, $a < 1$. The club is a dense near-clique of order $\sqrt{n}$ nodes without opening, and opening spreads it out. Between 0.83 and 1 the network has triangles but no hubs to speak of.

Condensation requires the census rule. When the triad is found locally, a random node introducing two friends or befriending a friend of a friend, the network stays scale-free and clustered with $\gamma = 1 + 2/(eu)$ at $\beta_- = 0$. The friend-of-a-friend version at $\beta_+ = \alpha$ is the faithful generalization of $LS(n, 1)$: same density, same exponent $\gamma = 5$. Local closure also builds a hierarchy of assortative communities. Census closure replaces it with a club and a periphery that barely link to each other, an assortativity of about 0.6 that persists into the hub-free phase.

The condensate differs from Bose–Einstein condensation in fitness models [9] and from superlinear preferential attachment [12]. There a single node captures a finite fraction of the links. Here a growing set of nodes does, every one of them with a degree that diverges with n. Without opening the set is a dense block of $\Theta(\sqrt{n})$ nodes, a sparse graph with a dense spike in graphon language.

Open questions. (i) A controlled theory of the onset $n^*(u)$, power law versus essential singularity, and of its seed-to-seed distribution. (ii) Whether the condensate boundary at $\beta_-/\beta_+ \approx 0.83$ is a sharp line or drifts with n, and a theory of the club exponent $a^*(\beta_-/\beta_+)$. (iii) The degree distribution of the tree phase, which the k^2 erosion argument should give in closed form. (iv) The scaling of the sparse core of rule (p), and its phase diagram with opening. (v) Mixtures of local and census closure, which should move both the onset and the boundary continuously.

Reproducibility. All results come from `tc.c` in the `triadic_closure` folder of the code repository (modes `l`, `w`, `p`, `c`, `f` and `r` for per-triad rates; the optional sixth argument is u_-). The scans are `scan.py`, `runs_ts.sh`, `runs_onset.sh` and `onset.py` (no opening),

and `scan_rm.py`, `runs_rm_ts.sh`, `club.py`, `cross.py` and `phase.py` (with opening), and `comm.py` (communities, run with graph-tool). `mkfigs.py` draws the figures. Summaries are in `data/`.

ACKNOWLEDGMENTS

All code is available at https://github.com/av2atgh/LocalNetworkGrowth. The calculations were assisted by Claude Opus 5, an AI assistant developed by Anthropic. Nodes & Links Ltd provided support in the form of salary for Alexei Vazquez but did not have any additional role in the conceptualization of the study, the analysis, the decision to publish or the preparation of the manuscript.

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